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First Report

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Participants

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First Visit to Bath: January 20th to February 2nd, 2019.

During this period I worked with Prof. Paul Milewski and Dr. Carlos Galeano Rios, a collaborator of Prof. Milewski. We started a joint work on the motion of a liquid droplet sustained on the surface of a vibrating fluid bath. I have been the responsible for modelling the free dynamics of the droplet as well as writting and integral expression for the force upon the droplet while in contact with the bath. Paul and Carlos will numerically compute the dynamics of the fluid in the bath and to integrate the equations for the motion of the droplet. The following sections contain a summary of the work I did during the visit.

1. Vertical motion of a drop bouncing on a fluid reservoir: The motion of the center of mass

We consider the three-dimensional vertical motion of a drop over the free-surface a fluid bath. The fluid in the drop and in the bath are incompressible with constant density ρ. The fluid inside the drop has viscosity μ and kinematic viscosity $\nu = \mu/\rho$. The density and the viscosity of the air are neglected and σ denotes the surface tension coefficient of the fluid-air interface.

The volume occupied by the drop at time $t \geq 0$ is denoted by $\Omega(t)$ that is supposed to be symmetric with respect to the vertical axis for all time. The fluid inside the drop satisfies the Navier-Stokes equations with surface tension

$$\begin{aligned}
&\mathbf{u}_t + \mathbf{u}\cdot\boldsymbol{\nabla}\mathbf{u} = -\frac{\boldsymbol{\nabla} p}{\rho} - g\boldsymbol{\nabla} z + \nu\Delta\mathbf{u} && \\
&\boldsymbol{\nabla}\cdot\mathbf{u} = 0 && t \geq 0,\ (x,y,z)\in\Omega(t)
\end{aligned} \tag{1}$$

$$p - \mu\big(\mathbf{n}\cdot(\boldsymbol{\nabla}\mathbf{u} + \boldsymbol{\nabla}\mathbf{u}^T)\mathbf{n}\big) = p_b + \sigma K \quad t \geq 0,\ (x,y,z)\in\partial\Omega(t), \tag{2}$$

$$\mu\big(\mathbf{n}\cdot(\boldsymbol{\nabla}\mathbf{u} + \boldsymbol{\nabla}\mathbf{u}^T)\boldsymbol{\tau}\big) = 0 \quad t \geq 0,\ (x,y,z)\in\partial\Omega(t), \tag{3}$$

where: g is the acceleration of gravity, p_b is the pressure due to the contact, if any, with the fluid bath, $\partial\Omega(t)$ is the boundary of $\Omega(t)$, K is twice the mean curvature of $\partial\Omega(t)$ at a given point, $\mathbf{n}$ is a unit vector normal to the boundary (pointing outward $\Omega(t)$), and $\boldsymbol{\tau}$ is any unit vector tangent to the boundary. The free-stress boundary condition is always assumed, even when there is contact of the drop and the fluid in the bath. In this case a film of air may lubrificate the interface between the drop and the bath and since the viscosity of air has been neglected there is no shear stress on the surface neither of the drop nor of the bath. The reference frame (x,y,z) is such that the center of mass of

$\Omega(t)$ lies on the vertical axis ($x = y = 0$), z increases upward, and $z = 0$ at the undisturbed surface of the bath. The unit vectors parallel to the coordinate lines are $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$.

The z coordinate of the center of mass of $\Omega(t)$ is given by

$$\overline{z}(t) = \frac{1}{V}\int_{\Omega(t)} z dV = \frac{1}{V}\int_{\Omega(0)} \Psi_3(t, x_0, y_0, z_0) dV_0,$$

where: $V = \int_{\Omega(t)} dV$ is the volume of the drop, $\Psi_3(t, x_0, y_0, z_0)$ is the z-coordinate of the fluid particle at time t that at time $t = 0$ was in (x_0, y_0, z_0), and $dV = dxdydz$ and $dV_0 = dx_0dy_0dz_0$ are the volume elements in the physical space and in the space of initial conditions, respectively. The identity

$$\ddot{\Psi}_3(t, x_0, y_0, z_0) = (\mathbf{u}_t + \mathbf{u}\cdot\boldsymbol{\nabla}\mathbf{u})\cdot\mathbf{e}_3,$$

where the right side is evaluated at (t, x, y, z), and Navier-Stokes equation imply

$$\ddot{\overline{z}}(t) = -\frac{g}{V}\int_{\Omega(t)} dV - \frac{1}{\rho V}\int_{\Omega(t)} (\boldsymbol{\nabla} p\cdot\mathbf{e}_3) dV = -g - \frac{1}{\rho V}\int_{\partial\Omega(t)} p(\mathbf{n}\cdot\mathbf{e}_3) dA$$

where $\rho V = M$ is the mass of the drop, dA is the surface element on $\partial\Omega(t)$, and the term that contains the integral of $\mathbf{e}_3\cdot(\Delta\mathbf{u})$ was omitted because the boundary condition (3) immediately implies it is null. The boundary condition (2) implies

$$\ddot{\overline{z}}(t) = -g - \frac{1}{M}\int_{\partial\Omega(t)} p_b(\mathbf{n}\cdot\mathbf{e}_3) dA - \sigma\frac{1}{M}\int_{\partial\Omega(t)} K(\mathbf{n}\cdot\mathbf{e}_3) dA \tag{4}$$

The last integral in the right side of the equation is zero because it represents the time derivative of the rigid translation of the surface $\partial\Omega(t)$ along the z-axis (a consequence of the "first variation of area formula" in differential geometry). Therefore the acceleration of the center of mass of the drop is given by

$$\ddot{\overline{z}}(t) = -g + \frac{F(t)}{M}, \quad \text{where} \quad F(t) = -\int_{\partial\Omega(t)} p_b(\mathbf{n}\cdot\mathbf{e}_3)dA \tag{5}$$

If the drop is not in contact with the free surface then the acceleration of its center of mass is equal to the acceleration due to gravity.

2. Navier-Stokes equation on the reference frame of the center of mass and the free motion of the drop

In order to describe the deformation of the drop it is convenient to write the Navier-Stokes equation on a reference frame $\{X, Y, Z\} = \{x, y, z - \overline{z}(t)\}$ that moves with the center of mass of the drop. In the new coordinates the velocity field becomes $\mathbf{U}(t, X, Y, Z) = \mathbf{u}(t, x, y, z) - \dot{\overline{z}}(t)$ and Navier-Stokes equations (1) with the boundary conditions (2) and (3) become

$$\begin{aligned} \mathbf{U}_t + \mathbf{U}\cdot\boldsymbol{\nabla}\mathbf{U} &= -\frac{\boldsymbol{\nabla}P}{\rho} - (g + \ddot{\overline{z}})\boldsymbol{\nabla}Z + \nu\Delta\mathbf{U} \\ &= -\frac{\boldsymbol{\nabla}P}{\rho} - \frac{F}{M}\boldsymbol{\nabla}Z + \nu\Delta\mathbf{U} \\ \boldsymbol{\nabla}\cdot\mathbf{U} &= 0 \end{aligned} \tag{6}$$

$$P - \mu(\mathbf{n}\cdot(\boldsymbol{\nabla}\mathbf{U} + \boldsymbol{\nabla}\mathbf{U}^T)\mathbf{n}) = P_b + \sigma K \tag{7}$$

$$\mu(\mathbf{n}\cdot(\boldsymbol{\nabla}\mathbf{U} + \boldsymbol{\nabla}\mathbf{U}^T)\boldsymbol{\tau}) = 0 \tag{8}$$

where $F(t)$, given in equation (5), is the z-component of the force that acts upon the drop when it is in contact with fluid bath and

$$P(t, X, Y, Z) = p(t, X, Y, Z + \overline{z}(t)) \quad \text{and} \quad P_b(t, X, Y, Z) = p_b(t, X, Y, Z + \overline{z}(t))$$

are the pressures written in the new coordinates.

Let (r, θ, ϕ) be the usual spherical coordinates in the frame (X, Y, Z), namely

$$X = r \sin\theta \cos\phi, \quad Y = r \sin\theta \sin\phi, \quad Z = r \cos\theta,$$

with coordinate unit vectors $(\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_\phi)$. Suppose that at all times the shape of the drop can be written as

$$r = R + \zeta(t, \theta) \tag{9}$$

where R is the radius of a spherical drop with the same volume (or the radius of the "undisturbed" drop). Notice that ζ does not depend on ϕ because the shape of the drop was assumed to be z-axisymmetric. In this paper it is further assumed that at all times ζ and its derivatives are small quantities such that all terms proportional to ζ^2 and its derivatives can be neglected. Under the small deformation hypothesis the free motion of the drop can be easily described. In this case $F(t) = 0$ and the deformation of the drop is determined by the linearized Navier-Stokes equation without gravity.

Following the book by Lamb, the deformation function ζ can be spatially decomposed into spherical harmonics

$$\zeta(t, \theta) = \sum_{l \geq 2} A_l(t) P_l(\cos\theta) \tag{10}$$

where P_l is the Legendre polynomial of order l and $A_l(t)$ is the time dependent amplitude of the mode l. Notice that the term $l = 0$ does not appear in the sum because the fluid inside the drop is incompressible, and neither the term $l = 1$ because the center of mass corresponds to $r = 0$. The analysis of the time evolution of A_l is simpler when the viscosity is null. Both cases $\mu = 0$ and $\mu > 0$ will be considered.

2.1. **Free motion with $\mu = 0$.** In this case equations (6) becomes Euler's equation without gravity. In this case the free motion dynamics of the coefficients A_l in equation (10) are given by (Lamb paragraph 275):

$$A_l(t) = a_l \cos(\omega_l t) + b_l \sin(\omega_l t) \tag{11}$$

where: a_l and b_l are constants and ω_l is the eigenfrequency of oscillation associated to the mode l

$$\omega_l^2 = l(l-1)(l+2)\frac{\sigma}{\rho R^3} \tag{12}$$

The constants a_l and b_l are determined from the initial configuration of the drop and its derivative by means of

$$\begin{aligned} a_l &= \frac{2l+1}{2}\int_0^\pi P_l(\cos\theta)\zeta(0,\theta)\sin\theta d\theta \\ b_l &= \frac{2l+1}{2\,\omega_l}\int_0^\pi P_l(\cos\theta)\zeta_t(0,\theta)\sin\theta d\theta, \end{aligned} \tag{13}$$

where the orthogonality relations of the Legendre polynomials were used:

(14)

$$\int_0^\pi P_l(\cos\theta)P_n(\cos\theta)\sin\theta d\theta = \int_{-1}^1 P_l(s)P_n(s)ds = \begin{cases} 0 & \text{if } l \neq n \\ \frac{2}{2l+1} & \text{if } l = n \end{cases}$$

Notice that in this case the dynamics of A_l is characterized by the differential equation

$$\ddot{A}_l = -\omega_l^2 A_l \tag{15}$$

2.2. **Free motion with $\mu > 0$ small.** If the viscosity $\mu > 0$ is small and the solution to the Navier-Stokes equation $\mathbf{U}$ is well approximated by a irrotational velocity field $\mathbf{U} \approx \boldsymbol{\nabla}\phi$, then there is a simple way to describe the decay of the free oscillations of $\mathbf{U}$ (Lamb paragraph 355).

The method is based on the formula for the dissipation of energy when $\mathbf{u} = \boldsymbol{\nabla}\phi$ (see Lamb paragraph 329 Eq. (13))

$$\dot{E} = -\frac{\mu}{2}\int_{\Omega(t)} \sum_{ij} \left(\partial_{x_i} u_j + \partial_{x_j} u_i\right)^2 dV = -\mu \int_{\Omega(t)} \Delta\left(|\boldsymbol{\nabla}\phi|^2\right) dV$$

Using that each $l \geq 2$ is associated to a velocity potential $\Phi(t, r, \theta) = \dot{A}_l(t)\frac{R}{l}\frac{r^l}{R^l}P_l(\cos\theta)$ (see equation (27)) an integration over a ball (Lamb paragraph 355) gives

$$\dot{E} \approx 2\frac{(l-1)(2l+1)}{l}\mu R \dot{A}_l^2 \frac{4\pi}{2l+1} = 2\tilde{F}_l$$

where $\tilde{F}_l$ is the so called Rayleigh dissipation function. The kinetic energy associated to this mode is

$$T_l = \frac{\rho}{2}\int_{\Omega(t)} |\boldsymbol{\nabla}\Phi|^2 dV \approx \frac{\dot{A}_l^2}{2}\frac{\rho R^3}{l}\frac{4\pi}{2l+1}$$

If $\mu = 0$ then the equation (15) for the oscillations, $\ddot{A}_l = -\omega_l^2 A_l$, can be written as

$$\frac{d}{dt}\frac{\partial T_l}{\partial \dot{A}_l} + \frac{\partial V_l}{\partial A_l} = 0$$

where V_l is a potential function for the mode l. This last equation and equation (15) imply

$$V_l = \omega_l^2 \frac{A_l^2}{2}\frac{\rho R^3}{l}\frac{4\pi}{2l+1}$$

The damped equations of motion can be written as:

$$\frac{d}{dt}\frac{\partial T_l}{\partial \dot{A}_l} + \frac{\partial V_l}{\partial A_l} + \frac{\partial \tilde{F}_l}{\partial \dot{A}_l} = 0$$

The substitution of the expressions for T_l, V_l, and $\tilde{F}_l$ into this equation gives

$$\ddot{A}_l = -\omega_l^2 A_l - 2\tilde{\lambda}_l \dot{A}_l, \quad \text{where} \quad \tilde{\lambda}_l = (l-1)(2l+1)\frac{\nu}{R^2} \tag{16}$$

2.3. **Free motion with $\mu > 0$.** In this case equations (6) become the linearized Navier-Stokes equation without gravity. The time-dependence of the coefficients A_l, $l \geq 2$, are determined under the assumption that $A_l(t)$ is proportional to $\exp(-\tilde{\sigma}t)$ (REID, W. H. 1960 "The oscillations of a viscous liquid drop" Quart. Appl. Math. 18, 86-89). The constant $\tilde{\sigma}$, possibly complex, must satisfy a characteristic equation given in Reid with the following notation. For a given $l \geq 2$, let

$$q^2 = \tilde{\sigma}\frac{R^2}{\nu}$$

and

$$\alpha^2 = \frac{\omega_l R^2}{\nu}$$

where ω_l is the frequency in equation (12) (for $\nu = 0$). Then q is a solution to the equation:

$$\begin{aligned} &\frac{\alpha^4}{q^4} + 1 = \frac{2(l-1)}{q^2}\left[l + (l+1)\frac{q - 2lQ_{l+1/2}(q)}{q - 2Q_{l+1/2}(q)}\right] \\ &\text{where} \qquad Q_{l+1/2}(q) = \frac{J_{l+3/2}(q)}{J_{l+1/2}(q)} \end{aligned} \tag{17}$$

and the J's are spherical Bessel functions. According to Reid, "this characteristic equations has solutions of two types. For values of α^2 which exceed a certain critical value, damped oscillations will occur, but for values of α^2 less than this critical value, two aperiodic modes of decay appear. The solution of Eq. (17) at this critical point for the principal mode $l = 2$ is (Chandrasehkar 1959)

$$\omega_2\frac{R^2}{\nu} = 3.69 \qquad \left(\text{or} \quad R_{crit} = \frac{3.69^2\mu^2}{8\sigma\rho}\right)$$

For a drop of water surrounded by air ($\sigma = 74$ dynes/cm) this gives a radius R = 0.23mm (this number seems to be wrong, I got 2.3×10^{-5} mm). Drops larger than this critical radius will therefore tend to oscillate while smaller drops will be aperiodically damped.” (copied from Reid except the equation for R_{crit})

So, if for a given $l \geq 2$ the characteristic values $\tilde{\sigma}$ are $\sigma_{al} = \lambda_l + i\,\omega_{\nu,l}$ and $\sigma_{bl} = \lambda_l - i\,\omega_{\nu,l}$ then

$$A_l(t) = \exp(-\lambda_l t)\Big(a_l \cos(\omega_{\nu,l} t) + b_l \sin(\omega_{\nu,l} t)\Big) \tag{18}$$

If the characteristic values $\tilde{\sigma}$ are $\sigma_{al} = \lambda_{al}$ and $\sigma_{bl} = \lambda_{bl}$ then

$$A_l(t) = a_l \exp(-\lambda_{al} t) + b_l \exp(-\lambda_{bl} t) \tag{19}$$

where in both equations (18) and (19) a_l and b_l are real constants.

As stated in Reid, if ν is small then $\tilde{\sigma} \approx \tilde{\lambda}_l \pm i\omega_l$ where $\tilde{\lambda}_l$ is the damping coefficient in equation (16) and ω_l is the frequency of oscillations with no viscosity.

Notice that in this case the dynamics of A_l is characterized by the differential equation

$$\ddot{A}_l = -\sigma_{al}\sigma_{bl} A_l - (\sigma_{al} + \sigma_{bl})\dot{A}_l \tag{20}$$

3. The motion of the drop while in contact with the fluid bath

A drop released from a sufficiently high point at $t = 0$ will take some time t_1 to first touch the surface of the fluid bath. Then for some time interval, $t_1 \leq t \leq t_2$, the drop will be over the pressure field P_b due to the fluid in the bath. This pressure field will act upon the motion of the center of mass and upon the configuration of the drop. The goal in this section is to obtain differential equations within the time interval $[t_1, t_2]$ for the coefficients $A_l(t)$, $l = 2, 3, \ldots$, that the determine the configuration of the drop by means of equation (10). The position of the center of mass of the drop, $\overline{z}(t)$, is determined by equation (5).

3.1. **Geometric preliminaries and another expression for $F(t)$.** The unit normal vector to the boundary of the drop (pointing outside) is the normalized gradient of the function $f(r,\theta,\phi) = r - \zeta(t,\theta)$ at the level set $f = R$. If ζ is decomposed into spherical harmonics as in equation (10) then

$$\boldsymbol{\nabla} f = f_r \mathbf{e}_r + \frac{1}{r} f_\theta \mathbf{e}_\theta + \frac{1}{r\sin\phi} f_\phi \mathbf{e}_\phi = \mathbf{e}_r + \frac{\mathbf{e}_\theta}{r} \sum_{l\geq 2} A_l P'(\cos\theta)\sin\theta$$

where the prime indicates differentiation with respect to the argument $s = \cos\theta$ of $P_l(s)$. This expression implies that $|\boldsymbol{\nabla} f| = 1$ up to terms of the order of $|\boldsymbol{\nabla}\zeta|^2$, so, up to terms of the same order, the normal field in a neighborhood of $\partial\Omega(t)$ is

$$\mathbf{n}(r,\theta) = \mathbf{e}_r + \frac{\mathbf{e}_\theta}{r} \sum_{l\geq 2} A_l P_l'(\cos\theta)\sin\theta \tag{21}$$

and the normal field over $\partial\Omega(t)$ is

$$\mathbf{n}(\theta) = \mathbf{e}_r + \frac{\mathbf{e}_\theta}{R} \sum_{l\geq 2} A_l P_l'(\cos\theta)\sin\theta \tag{22}$$

The element of area on a plane orthogonal to $\mathbf{n}$ is, up to terms of the order of $|\zeta|^2$, $dA = r^2\sin\theta d\theta d\phi$. Since $r = R+\zeta$ on $\partial\Omega(t)$, the element of area on the surface of the drop can be written as

$$dA = \left(R^2 + 2R\sum_{l\geq 2} A_l P_l(\cos\theta)\right)\sin\theta d\theta d\phi, \tag{23}$$

where terms of the order of $|\zeta|^2$ were neglected.

The equation (5) for the motion of the center of mass depends only on the geometry of the drop, namely on the coefficients A_l. Up to first order in ζ equation (5) can be written in the following way.[1] Let $P_b(t,\theta)$ be the polar angle parameterization of the pressure on $\partial\Omega(t)$, namely

$$P_b(t,\theta) = P_b(t,X,Y,Z) = P_b(t, r\sin\theta, 0, r\cos\theta) \quad \text{where}$$
$$r = R + \zeta(t,\theta) = R + \sum_{l\geq 2} A_l(t) P_l(\cos\theta)$$

The spherical harmonic decomposition of P_b is given by

$$\begin{aligned} P_b(t,\theta) &= \sum_{l\geq 0} B_l(t) P_l(\cos\theta), \quad \text{where} \\ B_l(t) &= \frac{2l+1}{2}\int_0^\pi P_l(\cos\theta) P_b(t,\theta)\sin\theta d\theta \end{aligned} \tag{24}$$

(where the letter B_l stands for "bath"). Equations (21)and (23) imply

$$F(t) = -\int_{\partial\Omega(t)} P_b(\mathbf{n}\cdot\mathbf{e}_3)dA = -2\pi R^2\int_0^\pi P_b\cos\theta\sin\theta d\theta$$
$$+ 2\pi R\int_0^\pi P_b\Big(\sum_{l\geq 2} A_l\big(\sin^2\theta P_l'(\cos\theta) - 2\cos\theta P_l(\cos\theta)\big)\Big)\sin\theta d\theta.$$

This expression can be simplified with the use of $P_1(s) = s$, where $s = \cos\theta$, and

$$(1-s^2)P_l'(s) - 2sP_l(s) = \frac{1}{2l+1}\big(l(l-1)P_{l-1}(s) - (l+2)(l+1)P_{l+1}(s)\big)$$

(see Arfken pg 750 equations (12.26) and (12.27)). So, up to the precision of the small deformation hypothesis, the force $F(t)$ can be written as

$$F(t) = -\frac{4\pi R^2}{3}B_1(t) + 4\pi R\sum_{l\geq 2}\frac{A_l}{2l+1}\left(\frac{l(l-1)}{2l-1}B_{l-1}(t) - \frac{(l+2)(l+1)}{2l+3}B_{l+1}(t)\right) \tag{25}$$

[1] As pointed out by Carlos, it is easier to compute the integral in equation (5) then to apply the formula obtained in this paragraph. Nevertheless the harmonic coefficients $B_1, B_2, \ldots$ have to be computed anyway. So, maybe, the computation of the force by the above sum is faster.

The first term in the right side of this equation does not depend explicitly on the deformation variables, it is the only term in the sum when the drop is spherical.

3.2. **The equations for the deformation variables, $\mu = 0$.** The derivation of the equations for the deformation variables A_l, $l \geq 2$, requires the same hypotheses used to obtain the normal modes of free oscillations: small deformations and potential flow inside the drop. This implies that Euler's equation (6) can be linearized and U can be written as the gradient of a potential function $\Phi(t, r, \theta)$. As a result, equation (6) can be integrated inside $\Omega(t)$ to give

$$\Phi_t + \frac{P}{\rho} + Z\frac{F}{M} = c \tag{26}$$

where c depends only on t. The kinematic condition $\boldsymbol{\nabla}\Phi = \dot{\zeta}$ on the surface of the drop and the incompressibility of the fluid, which implies $\Delta\Phi = 0$, determine

$$\Phi(t, r, \theta) = \sum_{l\geq 2} \dot{A}_l(t)\frac{R}{l}H_l(r, \theta), \quad \text{where} \quad H_l(r, \theta) = \frac{r^l}{R^l}P_l(\cos\theta) \tag{27}$$

are solid spherical harmonics. Indeed, $\boldsymbol{\nabla}\Phi \cdot \mathbf{e}_r = \sum_{l\geq 2} \dot{A}_l \frac{r^{l-1}}{R^{l-1}} P_l(\cos\theta)$ and at the surface of the drop $r = R + \zeta = R + \sum_{l\geq 2} A_l P_l(\cos\theta)$, so $\boldsymbol{\nabla}\Phi \cdot \mathbf{n} = \dot{\zeta}$ up to terms of the order of ζ^2.

The boundary condition (7) can be used to eliminate P from equation (26)

$$\Phi_t + \frac{P_b + \sigma K}{\rho} + Z\frac{F}{M} = c$$

Multiply both sides of this equation by $\boldsymbol{\nabla} H_l \cdot \mathbf{n}$ and then integrated over $\partial\Omega(t)$ to obtain

$$
\begin{aligned}
\int_{\partial\Omega(t)} \Phi_t(\boldsymbol{\nabla} H_l \cdot \mathbf{n}) dA &= -\frac{1}{\rho}\int_{\partial\Omega(t)} P_b(\boldsymbol{\nabla} H_l \cdot \mathbf{n}) dA \\
&- \frac{\sigma}{\rho}\int_{\partial\Omega(t)} K(\boldsymbol{\nabla} H_l \cdot \mathbf{n}) dA - \frac{F}{M}\int_{\partial\Omega(t)} Z(\boldsymbol{\nabla} H_l \cdot \mathbf{n}) dA
\end{aligned} \tag{28}
$$

(the integral of the constant term is zero because H_l is harmonic). Equation (22) implies that

$$
\boldsymbol{\nabla} H_l \cdot \mathbf{n} = \frac{l}{R} P_l(\cos\theta)
$$

and equation (23) implies

$$
(\boldsymbol{\nabla} H_l \cdot \mathbf{n}) dA = lRP_l(\cos\theta)\sin\theta d\theta d\phi \tag{29}
$$

where in both equations terms of the order of ζ were neglected. The expression for Φ_t in equation (27) implies that on $\partial\Omega(t)$

$$
\begin{aligned}
\Phi_t(t, R+\zeta(t,\theta),\theta) &= \sum_{k\geq 2} \ddot{A}_k(t)\frac{R}{k}\frac{(R+\zeta)^k}{R^k} P_k(\cos\theta) \\
&= \sum_{k\geq 2} \ddot{A}_k(t)\frac{R}{k} P_k(\cos\theta)\big(1+\mathcal{R}_1\big)
\end{aligned}
$$

where $\mathcal{R}_1$ is a term of the order of ζ. This equation and equation (29) imply

$$
\begin{aligned}
\int_{\partial\Omega(t)} \Phi_t(\boldsymbol{\nabla} H_l \cdot \mathbf{n}) dA &= lR^2 2\pi \sum_{k\geq 2}\frac{\ddot{A}_k}{k}\int_{-1}^{1} P_l(s)P_k(s) ds (1+\mathcal{R}_2) \\
&= R^2 2\pi \frac{2}{2l+1}\ddot{A}_l(1+\mathcal{R}_2)
\end{aligned}
$$

where $\mathcal{R}_2$ is a term of the order of ζ.

At this point it seems natural to neglect the term $\ddot{A}_l\mathcal{R}_2$ since it is quantity of the order of ζ^2. It is interesting to analyze the consequences of this approximation, which eventually will be bone. Let G denote the result of the computation of the integrals in the right side of equation (28) that, in combination with the left side, gives

$$(30)\qquad R^2 2\pi \frac{2}{2l+1}\ddot{A}_l(1+\mathcal{R}_2) = G \quad \text{or} \quad R^2 2\pi \frac{2}{2l+1}\ddot{A}_l = G(1-\mathcal{R}_2)$$

The function G have terms that do not depend on ζ but only on the dominant spherical shape of the drop. Therefore the disregard of $\mathcal{R}_2$ at this point is consistent with the neglect of all terms of the order of ζ that will appear in the computations of the integrals in the right side of equation (28). After these considerations

$$(31)\qquad \int_{\partial\Omega(t)} \Phi_t(\boldsymbol{\nabla} H_l \cdot \mathbf{n})dA = R^2 2\pi \frac{2}{2l+1}\ddot{A}_l$$

where the factor $\mathcal{R}_2$ has been neglected.

The integrals for Z and P_b in the right side of equation (28) are, for $l \geq 2$ and up to corrections of the order of ζ:

$$(32)\qquad \int_{\partial\Omega(t)} Z(\boldsymbol{\nabla} H_l \cdot \mathbf{n})dA = 2\pi R^2 l \int_{-1}^{1} P_1(s)P_l(s)ds = 0$$

(33)

$$\int_{\partial\Omega(t)} P_b(\boldsymbol{\nabla} H_l \cdot \mathbf{n})dA = 2\pi R l \sum_{k\geq 0} B_k \int_{-1}^{1} P_k(s)P_l(s)ds = 2\pi R l \frac{2}{2l+1}B_l$$

The last integral that must be computed involves the mean curvature. The twice mean curvature K on $\partial\Omega(t)$ is given by the divergence of a field $\mathbf{n}$ of unit vectors such that $\mathbf{n}$ is orthogonal to $\partial\Omega(t)$. Such a vector field is given in equation (21), namely

$$\mathbf{n}(r,\theta) = \mathbf{e}_r + \frac{\mathbf{e}_\theta}{r}\sum_{l\geq 2} A_l P_l'(\cos\theta)\sin\theta$$

The divergence of this field is given by

$$\boldsymbol{\nabla}\cdot\mathbf{n}(t,r,\theta) = \frac{2}{r} - \frac{1}{r^2\sin(\theta)}\sum_{l\geq 2} A_l(t)\partial_\theta\Big(\sin\theta\,\partial_\theta\big(P_l(\cos\theta)\big)\Big),$$

The evaluation of this divergence on $\partial\Omega(t)$ gives

$$K(t,\theta) = \boldsymbol{\nabla}\cdot\mathbf{n}(t, R+\zeta(t,\theta),\theta) = \frac{2}{R} - \frac{2}{R^2}\sum_{l\geq 2} A_l(t)P_l(\cos\theta)$$
$$- \frac{1}{R^2\sin(\theta)}\sum_{l\geq 2} A_l(t)\partial_\theta\Big(\sin\theta\partial_\theta\big(P_l(\cos\theta)\big)\Big)$$
$$= \frac{2}{R} - \frac{1}{R^2}\sum_{l\geq 2} A_l(t)\Big(\frac{1}{\sin\theta}\partial_\theta\Big(\sin\theta\partial_\theta\big(P_l(\cos\theta)\big)\Big) + 2P_l(\cos\theta)\Big)$$

The Legendre equation

$$\frac{1}{\sin\theta}\partial_\theta\Big(\sin\theta\partial_\theta\big(P_l(\cos\theta)\big)\Big) + l(l+1)P_l(\cos\theta) = 0$$

implies

$$K(t,\theta) = \frac{2}{R} + \frac{1}{R^2}\sum_{l\geq 2}(l+2)(l-1)A_l(t)P_l(\cos\theta) \tag{34}$$

Notice that the term $P_1(\cos\theta)$ does not appear in the spherical harmonic expansion of K. This is related to the identity $\int_{\partial\Omega(t)} K(\mathbf{n}\cdot\mathbf{e}_3)dA = 0$, used in equation (4), that is valid regardless the hypothesis of small deformations.

The expression for the mean curvature in equation (34) has two terms: one represents the constant spherical part and the other the deformation that is of the order of ζ. According to the approximations that have already been done the term of order ζ should be neglected. Nevertheless this term will be preserved due to the relatively large coefficient in front of it. Indeed, for drops of small radius the surface tension dominates over the pressure. Notice that the dominant constant term of K integrates to zero, $\frac{2}{R}\int(\boldsymbol{\nabla}H_l\cdot\mathbf{n})dA = 0$, independently of any approximation, because H_l is a harmonic function. So, the term $\mathcal{R}_2$ in equation (30), of the order of ζ, that could give a contribution when multiplied by this constant curvature term does not.

As a consequence of equations (29) and (34), for $l \geq 2$, and up to corrections of the order of ζ:

$$\int_{\partial\Omega(t)} K(\boldsymbol{\nabla} H_l \cdot \mathbf{n}) dA = 2\pi \frac{l(l+2)(l-1)}{R} \frac{2}{2l+1} A_l \tag{35}$$

Finally, the combination of equations (28), (31), (32), (33), and (35) gives:

$$\begin{aligned} \ddot{A}_l &= -\frac{\sigma}{R^3 \rho} l\,(l+2)\,(l-1)\,A_l - \frac{l}{R\rho} B_l \\ &= -\omega_l^2 A_l - \frac{l}{R\rho} B_l \end{aligned} \tag{36}$$

where ω_l is the free frequency of oscillations given in equation (12) and, from equation (24),

$$B_l(t) = \frac{2l+1}{2} \int_0^\pi P_l(\cos\theta) P_b(t,\theta) \sin\theta d\theta$$

The conclusion is that the differential equation for the amplitude A_l is just the equations for the free oscillations (15) plus a forcing term that is equal to the projection of the pressure on the l spherical harmonic times some normalization.

3.3. **The equations for the deformation variables, $\mu > 0$ small.** In this case the same procedure followed in Section 2.2 to introduce dissipation of energy to the free oscillation modes gives

$$\begin{aligned} \ddot{A}_l &= -\frac{\sigma}{R^3 \rho} l\,(l+2)\,(l-1)\,A_l - 2(l-1)(2l+1)\frac{\nu}{R^2}\dot{A}_l - \frac{l}{R\rho} B_l \\ &= -\omega_l^2 A_l - 2\tilde{\lambda}_l \dot{A}_l - \frac{l}{R\rho} B_l \end{aligned} \tag{37}$$

3.4. **The equations for the deformation variables, $\mu > 0$.** As in the description of the free modes of oscillations, the equations to be considered in this section are the Navier-Stokes equations (6) linearized about $\mathbf{U} = 0$, namely

$$\begin{aligned}
&\mathbf{U}_t = -\frac{\boldsymbol{\nabla} P}{\rho} - \frac{F}{M}\boldsymbol{\nabla} Z + \nu \Delta \mathbf{U} \\
&\boldsymbol{\nabla} \cdot \mathbf{U} = 0 \\
&P - \mu(\mathbf{n} \cdot (\boldsymbol{\nabla}\mathbf{U} + \boldsymbol{\nabla}\mathbf{U}^T)\mathbf{n}) = P_b + \sigma K \\
&\mu(\mathbf{n} \cdot (\boldsymbol{\nabla}\mathbf{U} + \boldsymbol{\nabla}\mathbf{U}^T)\boldsymbol{\tau}) = 0
\end{aligned} \tag{38}$$

At first suppose that the drop is not in contact with the bath, $P_b = 0$ and $F = 0$. Then for a given $l \geq 2$, equations (38) have two linearly independent solutions of the form $\mathbf{U} = \dot{A}_l \mathbf{U}_l$, where $\mathbf{U}_l(X, Y, Z)$ is some special function and $A_l(t)$ is one of the two solutions to equation (20). These two solutions are related to the the spherical harmonic representation of the function ζ. If ζ is given by $\zeta(t, \theta) = A_l(t) P_l \cos\theta$, then the initial condition $\big(A_l(0), \dot{A}_l(0)\big) = (1, 0)$ corresponds to one solution and $\big(A_l(0), \dot{A}_l(0)\big) = (0, 1)$ to the other. At $r = R$ the function $\mathbf{U}_l$ satisfies $\mathbf{U}_l(R, \theta, \phi) = P_l(\cos\theta)$ such that $\Sigma_l \dot{A}_l(\mathbf{e}_r \cdot \mathbf{U}_l) = \dot{\zeta}$ is verified at the surface of the drop up to terms of the order of ζ^2. To each solution $\dot{A}_l \mathbf{U}_l$ corresponds a pressure function P that is a harmonic function (take the divergence of the first equation in (38)). The boundary condition in the third line of equation (38) implies that P is proportional to a solid harmonic of order l

$$P(t, r, \theta) = D_l(t) \frac{r^l}{R^l} P_l(\cos\theta) = D_l(t) H_l(r, \theta)$$

(D_l stands for 'drop'). This boundary condition also implies that D_l is some linear combination of A_l and $\dot{A}_l$. D_l depends on A_l because A_l appears in the spherical harmonics expansion of the mean curvature (equation (34)), and D_l depends on $\dot{A}_l$ because the normal viscous stress depends linearly on $\mathbf{U}$.

Using all these facts it is possible to obtain some information about equation (20) without having to solve the boundary value problem associated to equation (38). Consider the following decomposition of $\mathbf{U}_l$

$$\mathbf{U}_l(r,\theta) = \frac{R}{l}\boldsymbol{\nabla} H_l(r,\theta) + \big(\mathbf{U}_l(r,\theta) - \frac{R}{l}\boldsymbol{\nabla} H_l(r,\theta)\big) = \frac{R}{l}H_l(r,\theta) + \mathbf{V}(r,\theta)$$

where $H_l = (r/R)^l P_l(\cos\theta)$ is harmonic and for $r = R$

$$\mathbf{e}_r \cdot \mathbf{V}(R,\theta) = \left(\mathbf{e}_r \cdot \left(\mathbf{U}_l(R,\theta) - \frac{R}{l}\boldsymbol{\nabla} H_l(R,\theta)\right)\right) = 0$$

The substitution of $\mathbf{U} = \dot{A}_l \mathbf{U}_l = \dot{A}_l\big(\frac{R}{l}\boldsymbol{\nabla} H_l + \mathbf{V}_l\big)$ and $P = D_l H_l$ into the first line of equation (38) gives

$$\ddot{A}_l \frac{R}{l}\boldsymbol{\nabla} H_l + \ddot{A}_l \mathbf{V}_l = -D_l \frac{\boldsymbol{\nabla} H_l}{\rho} + \dot{A}_l \nu \Delta \mathbf{V}_l$$

where it was used that $\Delta\boldsymbol{\nabla} H_l = 0$. If this equation is multiplied by $\boldsymbol{\nabla} H_l$ and integrated over a ball of radius R, then

$$\begin{aligned} &\ddot{A}_l \frac{R}{l}\int (\boldsymbol{\nabla} H_l \cdot \boldsymbol{\nabla} H_l) dV + \ddot{A}_l \int (\boldsymbol{\nabla} H_l \cdot \mathbf{V}_l) dV \\ &= -\frac{D_l}{\rho}\int (\boldsymbol{\nabla} H_l \cdot \boldsymbol{\nabla} H_l) dV + \dot{A}_l \nu \int (\boldsymbol{\nabla} H_l \cdot \Delta \mathbf{V}_l) dV \end{aligned} \tag{39}$$

The integrals

$$\int (\boldsymbol{\nabla} H_l \cdot \mathbf{V}_l) dV = \int H_l (\mathbf{e}_r \cdot \mathbf{V}_l) dA = 0.$$

and

$$\int (\boldsymbol{\nabla} H_l \cdot \boldsymbol{\nabla} H_l) dV = \int H_l (\mathbf{e}_r \cdot \boldsymbol{\nabla} H_l) dA = l\, R\, \frac{4\pi}{2l+1}$$

can be used to rewrite equation (39) as

$$\ddot{A}_l = -D_l \frac{l}{R\rho} + \dot{A}_l \frac{\nu R^2 (2l+1)}{4\pi} \int (\boldsymbol{\nabla} H_l \cdot \Delta \mathbf{V}_l) dV \tag{40}$$

If $D_l(t)$ is written as a linear combination of $\dot{A}_l(t)$ and $A_l(t)$, then this equation becomes a second-order constant coefficient equation for A_l. The conclusion is that the two linearly independent solutions A_l

verify equation (40) and equation (20), so these two second order linear ordinary differential equations with constant coefficients must be the same:

$$
\begin{aligned}
\ddot{A}_l &= -D_l\frac{R}{l\rho} + \dot{A}_l\frac{\nu}{C}\int(\boldsymbol{\nabla}H_l\cdot\Delta\mathbf{V}_l)dV \\
&= -\sigma_{al}\sigma_{bl}A_l - (\sigma_{al}+\sigma_{bl})\dot{A}_l
\end{aligned} \tag{41}
$$

Now, consider the case where the drop is in contact with the bath, $P_b \neq 0$ and $F \neq 0$. The spherical harmonic decomposition of P_b over $\partial\Omega(t)$ is given in equation (24), namely $P_b(t,\theta) = \sum_{l\geq 2} B_l(t)P_l(\cos\theta)$. Let $\tilde{P}_b$ be a harmonic function inside $\Omega(t)$ that has the same values of P_b on $\partial\Omega(t)$. Up to terms of the order of ζ, $\tilde{P}_b$ is given by

$$\tilde{P}_b(t,r,\theta) = \sum_{l\geq 0} B_l(t)H_l(r,\theta)$$

Consider the change of variables

$$\tilde{P}(t,r,\theta) = P(t,r,\theta) - \tilde{P}_b(t,r,\theta)$$

In terms of the new pressure $\tilde{P}$ equation (38) becomes

$$
\begin{aligned}
&\mathbf{U}_t = -\frac{\boldsymbol{\nabla}\tilde{P}}{\rho} - \frac{F}{M}\boldsymbol{\nabla}Z + \nu\Delta\mathbf{U} - \frac{\boldsymbol{\nabla}\tilde{P}_b}{\rho} \\
&\boldsymbol{\nabla}\cdot\mathbf{U} = 0 \\
&\tilde{P} - \mu(\mathbf{n}\cdot(\boldsymbol{\nabla}\mathbf{U} + \boldsymbol{\nabla}\mathbf{U}^T)\mathbf{n}) = \sigma K \\
&\mu(\mathbf{n}\cdot(\boldsymbol{\nabla}\mathbf{U} + \boldsymbol{\nabla}\mathbf{U}^T)\boldsymbol{\tau}) = 0
\end{aligned} \tag{42}
$$

Notice that the boundary conditions are the same as those in the case of free oscillations, the difference is a new forcing term in the momentum equation. The solution to this new problem can be written as a linear combination

$$\mathbf{U}(t,r,\theta) = \sum_{l\geq 0} A_l(t)\mathbf{U}_l(r,\theta)$$

where $\mathbf{U}_l = \frac{R}{l}\boldsymbol{\nabla} H_l + \mathbf{V}_l$ are the eigenfunctions of the free modes of oscillations and A_l are time dependent coefficients to be determined. For $l \geq 2$, this sum can be substituted into the equation in the first line of (42), then multiplied by $\boldsymbol{\nabla} H_l$ and integrated over the ball. Neglecting terms of the order of ζ and using that $\int(\boldsymbol{\nabla} H_l \cdot \boldsymbol{\nabla} Z)dV = 0$, because $Z = rP_1(\cos\theta)$, the result is

$$\ddot{A}_l \frac{R}{l}\int(\boldsymbol{\nabla} H_l \cdot \boldsymbol{\nabla} H_l)dV = -\frac{\tilde{D}_l}{\rho}\int(\boldsymbol{\nabla} H_l \cdot \boldsymbol{\nabla} H_l)dV$$
$$+ \dot{A}_l \nu \int(\boldsymbol{\nabla} H_l \cdot \Delta \mathbf{V}_l)dV - \frac{B_l}{\rho}\int(\boldsymbol{\nabla} H_l \cdot \boldsymbol{\nabla} H_l)dV$$

that is similar to equation (39) except for the additional pressure term. In this equation $\tilde{D}_l$ represents the coefficient of spherical harmonic decomposition of $\tilde{P}$, it replaces the D_l in equation (39). Since the boundary condition $\tilde{P} - \mu(\mathbf{n} \cdot (\boldsymbol{\nabla}\mathbf{U} + \boldsymbol{\nabla}\mathbf{U}^T)\mathbf{n}) = \sigma K$ is the same as in the case of free oscillations the dependence of $\tilde{D}_l$ on the coefficients A_l and $\dot{A}_l$ is the same as in the case of free oscillations. Therefore the new equation for A_l is that for the free oscillations (41) plus the extra forcing term, namely

$$\ddot{A}_l = -\sigma_{al}\sigma_{bl}A_l - (\sigma_{al} + \sigma_{bl})\dot{A}_l - \frac{l}{R\rho}B_l \tag{43}$$

Notice that the forcing term due to P_b is the same that appears in equations (36) and (37) that represent the situations where $\nu = 0$ and $\nu > 0$ is small, respectively.

4. Summary

The equations for the motion of the drop for $\mu > 0$ are:

- The z coordinate of the center of mass of is

$$\overline{z}(t) = \frac{1}{V}\int_{\Omega(t)} z dV$$

where: $\Omega(t)$ is the region occupied by the drop and $V = \int_{\Omega(t)} dV$ is the volume of the drop. The equation (5) for the motion of $\overline{z}$ is

$$\ddot{\overline{z}}(t) = -g + \frac{F(t)}{M}, \quad \text{where} \quad F(t) = -\int_{\partial\Omega(t)} p_b(\mathbf{n}\cdot\mathbf{e}_3)dA,$$

where $M = \rho V$ is the total mass of the drop, and p_b is the pressure upon the drop due to the contact, if any, with the fluid bath.

- In a coordinate system parallel to the inertial one but centered at $\overline{z}$, $\{X, Y, Z\} = \{x, y, z - \overline{z}(t)\}$, the boundary of the drop is described as

$$r = R + \zeta(t,\theta) \quad \text{where} \quad \zeta(t,\theta) = \sum_{l\geq 2} A_l(t)P_l(\cos\theta)$$

where (r, θ, ϕ) are spherical coordinates with respect $\{X, Y, Z\}$ and P_l is the Legendre polynomial of order l.

- During the free flight of the drop the coefficients A_l, $l \geq 2$, are determined by (equation (20))

$$\ddot{A}_l = -\sigma_{al}\sigma_{bl}A_l - (\sigma_{al} + \sigma_{bl})\dot{A}_l$$

where σ_{al} and σ_{bl} are two roots of the characteristic equation (17) for $\tilde{\sigma}$. This characteristic equation is

$$\frac{\alpha^4}{q^4} + 1 = \frac{2(l-1)}{q^2}\left[l + (l+1)\frac{q - 2lQ_{l+1/2}(q)}{q - 2Q_{l+1/2}(q)}\right]$$

where:

$$Q_{l+1/2}(q) = \frac{J_{l+3/2}(q)}{J_{l+1/2}(q)},$$

the J's are spherical Bessel functions,

$$q^2 = \tilde{\sigma}\frac{R^2}{\nu}$$

and

$$\alpha^2 = \frac{\omega_l R^2}{\nu} \quad \text{with} \quad \omega_l^2 = l(l-1)(l+2)\frac{\sigma}{\rho R^3}$$

(ω_l is the frequency of oscillations when $\nu = 0$). Depending on ν and on the size of the drop the oscillations may be underdamped of overdamped (see equations (18) and (19)).

- During the time interval where the drop is in contact with the fluid bath the coefficients A_l, $l \geq 2$, are determined by (equation (43))

$$\ddot{A}_l = -\sigma_{al}\sigma_{bl}A_l - (\sigma_{al} + \sigma_{bl})\dot{A}_l - \frac{l}{R\rho}B_l$$

where (equation (24))

$$B_l(t) = \frac{2l+1}{2}\int_0^{\pi} P_l(\cos\theta)P_b(t,\theta)\sin\theta d\theta$$

In this last equation, notice that

$P_b(t,\theta) = P_b(t,X,Y,Z) = P_b(t, r\sin\theta, 0, r\cos\theta) \quad \text{where}$

$r = R + \zeta(t,\theta) = R + \sum_{l\geq 2} A_l(t)P_l(\cos\theta)$